# Gauss' Theorema Egregium

# An End-to-End Meta-Math Walkthrough

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Abstract

A complete authored walkthrough of gauss' theorema egregium, with explicit mathematical contracts, local verification gates, and reusable exports. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

**DG-03 / Differential Geometry / Report v1**

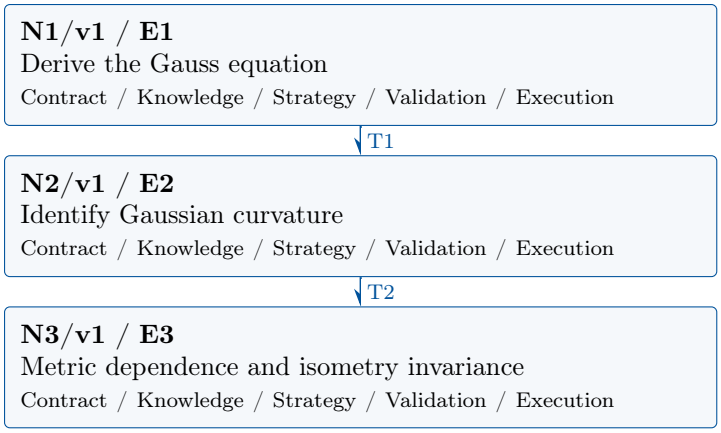
## I. TARGET AND IMPORT CONTRACT

**Theorem I.1.** *For a smooth regular surface in $\mathbb{R}^3$, prove Gaussian curvature $K = \det S$ is determined by the first fundamental form and is preserved by local isometries.*

### A. Foundations and permitted imports

For a regular surface choose local coordinates and a unit normal $\nu$. Use the induced metric g, its Levi-Civita connection $\nabla$, and decomposition of the ambient derivative $D_X Y = \nabla_X Y + h(X,Y)\nu$ with $D_X \nu = -SX$ and $h(X,Y) = g(SX,Y)$. These follow from orthogonal projection, metric compatibility, and torsion-freeness. Curvature convention $R(X,Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z$. The Gauss equation is derived here.

## II. NODE INTERACTION WORKFLOW




Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

### A. Versioned evidence-transfer ledger

**T1: N1/E1/v1 to N2.** Intrinsic curvature expressed through h with its exact signs.

**Receiving activity.** N2: Identify Gaussian curvature

**Compatibility.** Use the same mathematical objects and hypotheses as the source export. Receiving contract: Orthonormal tangent basis $e_1, e_2$; prior exports E1/v1

**T2: N2/E2/v1 to N3.** $K = \det S$ is the scalar computed from the intrinsic curvature tensor.

**Receiving activity.** N3: Metric dependence and isometry invariance

**Compatibility.** Use the same mathematical objects and hypotheses as the source export. Receiving contract: Coordinate curvature formula and metric compatibility; prior exports E2/v1

## III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

### A. N1/v1: Derive the Gauss equation

**Object and role.** Extract tangential terms from ambient curvature zero.

**Inputs.** Flat ambient connection and tangential/normal decomposition; foundational imports stated in §1

**C — Contract**

Own E1/v1: Intrinsic curvature expressed through h with its exact signs.

**K — Knowledge**

Retain Flat ambient connection and tangential/normal decomposition; foundational imports stated in §1; keep their assumptions and versions attached.

**S — Strategy**

Extract tangential terms from ambient curvature zero.

**V — Validation**

Changing the curvature convention changes formula signs.

**E — Execution**

Carry out the numbered deductions and publish E1/v1 together with its hypotheses and proof location.

**Numbered mathematical activities**

N1.1 In $R^3$, $D_X D_Y Z - D_Y D_X Z - D_{[X,Y]} Z = 0$.

N1.2 The tangential part of $D_X D_Y Z$ is $\nabla_X \nabla_Y Z - h(Y,Z)SX$; interchange X,Y and subtract.

N1.3 Therefore $R(X,Y)Z = h(Y,Z)SX - h(X,Z)SY$.

N1.4 Pairing with W gives $g(R(X,Y)Z,W) = h(Y,Z)h(X,W) - h(X,Z)h(Y,W)$.

N1.5 This is the Gauss equation with the fixed curvature convention.

**Export E1/v1.** Intrinsic curvature expressed through h with its exact signs.

**Acceptance gate.** Changing the curvature convention changes formula signs. Publish only after checking every displayed deduction.

**Targeted repair.** If the check fails, revise N1: Changing the curvature convention changes formula signs. Recheck dependent nodes N2, N3.

### B. N2/v1: Identify Gaussian curvature

**Object and role.** Compute sectional curvature as a two-by-two determinant.

**Inputs.** Orthonormal tangent basis $e_1, e_2$; prior exports E1/v1

**C — Contract**

Own E2/v1: $K = \det S$ is the scalar computed from the intrinsic curvature tensor.

**K — Knowledge**

Retain Orthonormal tangent basis $e_1, e_2$; prior exports E1/v1; keep their assumptions and versions attached.

**S — Strategy**

Compute sectional curvature as a two-by-two determinant.

**V — Validation**

Reversing $\nu$ negates S but preserves its two-dimensional determinant.

**E — Execution**

Carry out the numbered deductions and publish E2/v1 together with its hypotheses and proof location.

**Numbered mathematical activities**

N2.1 Use $X = e_1, Y = e_2, Z = e_2, W = e_1$.

N2.2 The equation gives $g(R(e_1, e_2)e_2, e_1) = h_{11}h_{22} - h_{12}^2 = \det S$, since h is symmetric and the basis orthonormal.

N2.3 For coordinate vectors, $K = g(R(\partial_1, \partial_2)\partial_2, \partial_1)/(g_{11}g_{22} - g_{12}^2)$.

N2.4 The denominator is positive by regularity.

N2.5 Hence Gaussian curvature equals the intrinsic sectional curvature.

**Export E2/v1.** $K = \det S$ is the scalar computed from the intrinsic curvature tensor.

**Acceptance gate.** Reversing $\nu$ negates S but preserves its two-dimensional determinant. Publish only after checking every displayed deduction.

**Targeted repair.** If the check fails, revise N2: Reversing $\nu$ negates S but preserves its two-dimensional determinant. Recheck dependent nodes N3.

### C. N3/v1: Metric dependence and isometry invariance

**Object and role.** Express the connection and curvature entirely through g.

**Inputs.** Coordinate curvature formula and metric compatibility; prior exports E2/v1

**C** **Contract**

Own E3/v1: Gaussian curvature is determined by the first fundamental form and preserved by local isometries.

**K** **Knowledge**

Retain Coordinate curvature formula and metric compatibility; prior exports E2/v1; keep their assumptions and versions attached.

**S** **Strategy**

Express the connection and curvature entirely through g.

**V** **Validation**

Equal extrinsic pictures are unnecessary; the connection is metric-determined.

**E** **Execution**

Carry out the numbered deductions and publish E3/v1 together with its hypotheses and proof location.

**Numbered mathematical activities**

N3.1 The Christoffel symbols are $\Gamma_{ij}^{k} = \frac{1}{2} g^{k\ell}(\partial_i g_{j\ell} + \partial_j g_{i\ell} - \partial_\ell g_{ij})$.

N3.2 In a coordinate frame $R(\partial_i, \partial_j)\partial_k$ has component $\partial_i \Gamma_{jk}^{\ell} - \partial_j \Gamma_{ik}^{\ell} + \Gamma_{jk}^{m}\Gamma_{im}^{\ell} - \Gamma_{ik}^{m}\Gamma_{jm}^{\ell}$.

N3.3 Thus the displayed K uses only g and its first and second derivatives.

N3.4 A local isometry preserves g and, by uniqueness of the Levi-Civita connection, transports $\nabla$ and R; the scalar K is consequently preserved.

N3.5 This proves the theorem.

**Export E3/v1.** Gaussian curvature is determined by the first fundamental form and preserved by local isometries.

**Acceptance gate.** Equal extrinsic pictures are unnecessary; the connection is metric-determined. Publish only after checking every displayed deduction.

**Targeted repair.** If the check fails, revise N3: Equal extrinsic pictures are unnecessary; the connection is metric-determined. Withhold this terminal result until its gate passes.

## IV. CONTINUOUS MATHEMATICAL PROOF

Read these steps in order as one proof; the diagram is optional. Numbered activities identify the specific constructions and deductions owned by each node.

### A. §4.N1 — Derive the Gauss equation

1. In $R^3$, $D_X D_Y Z - D_Y D_X Z - D_{[X,Y]} Z = 0$.
2. The tangential part of $D_X D_Y Z$ is$\nabla_X \nabla_Y Z - h(Y,Z)SX$; interchange X,Y and subtract.
3. Therefore $R(X,Y)Z = h(Y,Z)SX - h(X,Z)SY$.
4. Pairing with W gives $g(R(X,Y)Z,W) = h(Y,Z)h(X,W) - h(X,Z)h(Y,W)$.
5. This is the Gauss equation with the fixed curvature convention.

### B. §4.N2 — Identify Gaussian curvature

1. Use $X = e_1, Y = e_2, Z = e_2, W = e_1$.
2. The equation gives $g(R(e_1,e_2)e_2,e_1) = h_{11}h_{22} - h_{12}^2 = \det S$, since h is symmetric and the basis orthonormal.
3. For coordinate vectors, $K = g(R(\partial_1,\partial_2)\partial_2,\partial_1)/(g_{11}g_{22} - g_{12}^2)$.
4. The denominator is positive by regularity.
5. Hence Gaussian curvature equals the intrinsic sectional curvature.

### C. §4.N3 — Metric dependence and isometry invariance

1. The Christoffel symbols are $\Gamma_{ij}^k = \frac{1}{2} g^{k\ell}(\partial_i g_{j\ell} + \partial_j g_{i\ell} - \partial_\ell g_{ij})$.
2. In a coordinate frame $R(\partial_i,\partial_j)\partial_k$ has component $\partial_i \Gamma_{jk}^\ell - \partial_j \Gamma_{ik}^\ell + \Gamma_{jk}^m \Gamma_{im}^\ell - \Gamma_{ik}^m \Gamma_{jm}^\ell$.
3. Thus the displayed K uses only g and its first and second derivatives.
4. A local isometry preserves g and, by uniqueness of the Levi-Civita connection, transports $\nabla$ and R; the scalar K is consequently preserved.
5. This proves the theorem.

□

## V. WORKED ORCHESTRATION

A cylinder of radiusr is parametrized by (r cosu,r sinu,v) and has metric $r^2du^2 + dv^2$. Set $x = ru, y = v$; the metric becomes $dx^2 + dy^2$, so all $\Gamma$ and R vanish and $K = 0$. Its principal curvatures have one zero value, giving the same detS despite its visible bending.

### A. Boundary and hypothesis audit

A sphere has positive K whereas the plane has $K = 0$, so no local isometry can flatten a sphere into the plane. Validation rejects equating arbitrary bending with an isometry; the metric, not only the embedding, must be preserved.

## VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

### A. Failure and correction

This is an authored teaching trace, not a recorded agent run.

1. **Contract → Knowledge:** register the §1 theorem, imports, and requested exports before beginning the proof.
2. **Strategy → Execution:** choose the bridges in §3 and carry out the numbered activities in §4; transmit only established E-exports along the §2 DAG.
3. **Validation → responsible node:** test the specific boundary/failure in §5 and the node's individual audit. A failing candidate is withheld; downstream nodes may not use it.
4. **Knowledge + Strategy:** identify whether the error is an omitted condition, an invalid inference, or an attempted extension beyond the theorem. Restore the exact contract or explicitly state a different one.
5. **Execution:** redo the affected mathematical activity using the repaired condition or inference. Recheck consuming nodes, rather than feeding their conclusions back as premises.
6. **Contract:** publish only the supported export with its assumptions and proof location. A hypothetical revised export is v2; the written proof here is the v1 exposition, not a claim that this repair was executed.

### B. Hypothesis boundary

A sphere has positive K whereas the plane has $K = 0$, so no local isometry can flatten a sphere into the plane. Validation rejects equating arbitrary bending with an isometry; the metric, not only the embedding, must be preserved.

### C. Version and recheck

Treat any changed assumption as a new contract. Recheck the revised export and all of its descendants. A backward diagnostic never establishes an earlier theorem. This is an authored teaching trace, not an execution log.

**Provenance.** This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

**Further reading.**

- Harvard Math 230a: Riemannian geometry, curvature, and Hopf–Rinow.